Flexible Tactile Sensing and Hierarchical Neural Inversion System

From dynamic pressing signals to interpretable intraoperative nodule detection, size inversion, and coarse depth prediction

1. Intraoperative tactile acquisition

Flexible array: 12x8 tactile grid

Dynamic pressing sequence: T = 10, stride = 2

42 size-depth conditions x 3 repetitions

Ex vivo porcine lung with embedded nodule phantoms

2. Hierarchical learning pipeline

Stage 1 detector: test AUC = 0.8383

Unified inverter: size top1 = 0.6180

Unified inverter: hard-route depth bAcc = 0.5337

Route-aware optimization: GT / hard / soft / top2 route

3. Interpretable deployment

XGBoost depth baseline bAcc = 0.5138

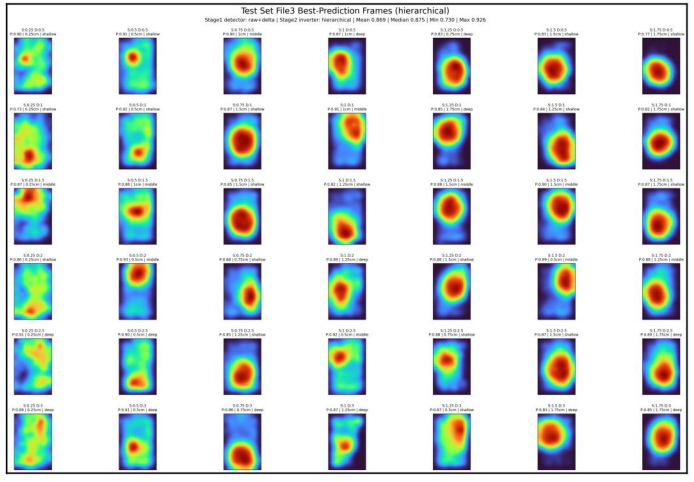
Latent probe + branch ablation + hard-pair

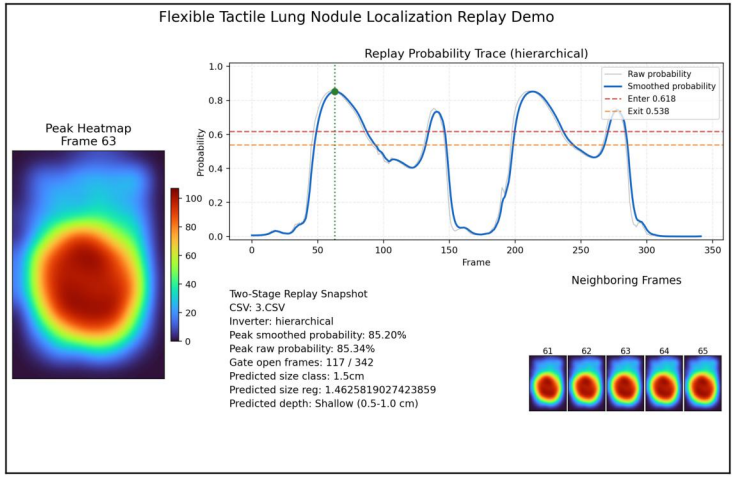
Phase occlusion validates temporal strategy

Real-time GUI: probability, size, depth, confidence

Clinical-to-algorithm overview

Test-set 6x7 condition matrix (best predicted frame per condition)



System outputs for paper and deployment

Real-time interface with detector + hierarchical inverter